

GAN-Based EEG Artifact Removal Using the TUH EEG Artifact Corpus (TUAR)

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Abstract

This project employs an LSTM-based Generative Adversarial Network (GAN) to remove artifacts from clinical EEG recordings using the TUAR dataset. Traditional EEG denoising methods often rely on simulated data, which limits their real-world utility. Our model addresses two primary challenges: robust removal of diverse, overlapping artifacts while preserving the underlying EEG signals, and overcoming the scarcity of artifact-free clinical recordings. By combining synthetic artifact generation with semi-supervised GAN training, our approach significantly enhances the quality of EEG data for clinical analysis. The system is implemented in PyTorch, adapting a validated LSTM-based GAN architecture, with performance evaluated through metrics of artifact reduction and signal preservation accuracy.

Keywords - GAN, LSTM, EEG, artifact, denoising, TUAR

1 Introduction

1.1 Context/Background

Electroencephalography (EEG) is essential for clinical diagnosis, neuroscience, and brain-computer interfaces due to its noninvasive capture of brain activity (Liu et al., 2021). However, EEG signals are frequently corrupted by artifacts such as ocular movements, muscle activity, chewing, and electrode malfunctions, which distort true neural signals and may lead to diagnostic errors (Fatourehchi et al., 2007; Zhang & Liu, 2021). Traditional methods—filtering, ICA, regression, and wavelet decomposition—often struggle with overlapping artifacts and typically require manual intervention (Chen & Wang, 2017; Jurcak et al., 2007).

1.2 Goals/Objectives

This project leverages the TUAR dataset (Corpus, 2023) to develop an LSTM-based GAN for artifact removal. Implemented in PyTorch using semi-supervised training and synthetic artifact generation informed by TUAR annotations, our approach aims to effectively remove diverse artifacts while preserving true EEG signals, thereby enhancing signal quality for clinical analysis (Liu et al., 2021; Zhang & Liu, 2021).

2 Related Work

2.1 Traditional Artifact Removal Methods

Traditional artifact removal techniques—including filtering, ICA, regression, and wavelet decomposition—have been applied to EEG data (Fatourehchi et al., 2007). Although ICA can decompose signals into independent components, it generally requires manual selection of artifact clusters and is less efficient when artifacts overlap with neural signals (Jurcak et al., 2007).

2.2 Deep Learning Methods

Deep learning methods have emerged as promising alternatives by capturing complex, non-linear data patterns (Liu et al., 2021). For example, dual-pathway CNN-LSTM models (Gao et al., 2021) and GAN-based frameworks like EEGANet (Sawangjai et al., 2021) have been shown to effectively remove artifacts. Nonetheless, many of these approaches are trained on synthetic or controlled datasets, which limits their generalizability to clinical EEG data.

2.3 Generative Adversarial Networks (GANs)

GANs offer a compelling solution by learning to generate realistic denoised signals. LSTM-based GANs have effectively removed simulated EEG artifacts (Brophy et al., 2022), and Wasserstein GANs (WGANs) have improved training stability and artifact removal performance (Dong et al., 2023). However, robust clinical validation on real-world datasets like TUAR remains a critical challenge (Corpus, 2023).

3 Problem Statement

This project tackles two primary challenges in EEG artifact removal:

1. **Artifact Removal with Signal Preservation:** Existing methods often fail to eliminate multiple types of artifacts without degrading the underlying EEG signals. An ideal solution must effectively remove artifacts while retaining essential neural activity for precise clinical analysis (Liu et al., 2021; Zhang & Liu, 2021).
2. **Lack of Artifact-Free Ground Truth:** Clinical EEG recordings rarely offer truly clean signals, posing significant difficulties for supervised approaches that depend on ground truth data. This situation necessitates the development of innovative, semi-supervised methods to reliably denoise EEG recordings (Brophy et al., 2022; Dong et al., 2023).

4 Solution

This project employs an LSTM-GAN for EEG artifact removal using the TUAR dataset. Our solution consists of four key parts: an overview of the LSTM-GAN architecture, data preprocessing with artifact masking, model training, and the inference workflow.

4.1 3.1 Overview of the LSTM-GAN Approach

The model consists of two main components:

1. **Generator (G):** A 2-layer stacked LSTM with 100 units per layer and $\tanh$ activation that learns to produce denoised EEG from noisy inputs (Brophy et al., 2022).
2. **Discriminator (D):** A 1D CNN that distinguishes between the generated denoised EEG and real clean EEG segments, thereby guiding the generator through adversarial training (Dong et al., 2023).

The TUAR dataset—comprising 310 bipolar EEG recordings (20–21 channels, nearly 100 hours total)—is segmented into 4-second windows with artifact annotations to create synthetic noisy/clean signal pairs. Figure 1 provides an overview of the TUAR dataset, and Figure 2 illustrates the LSTM-GAN architecture.

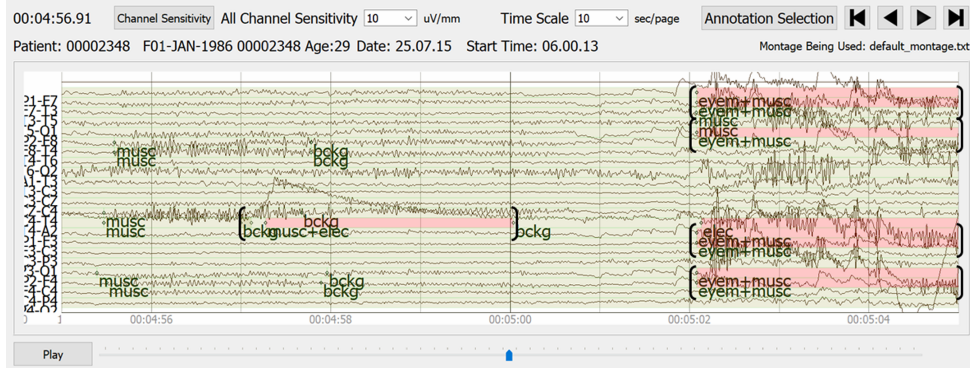


Figure 1: Overview of the TUAR dataset: summary of EEG recordings, channels, and annotated artifacts.

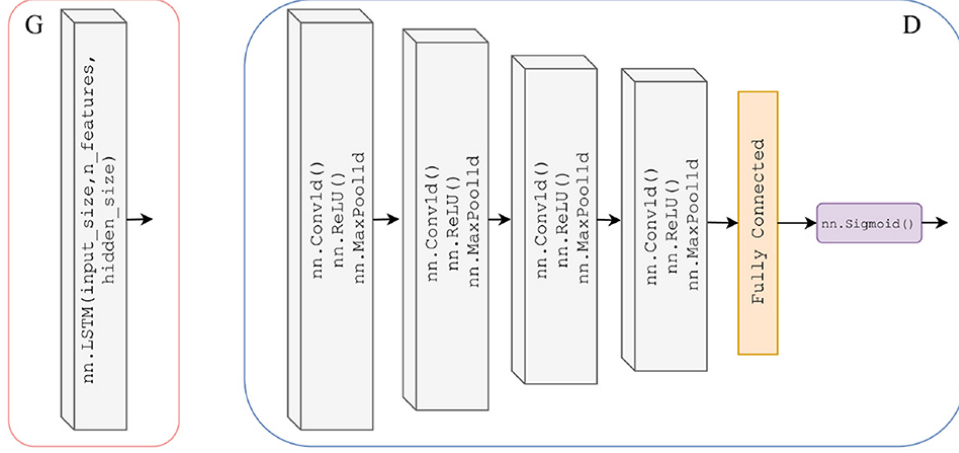


Figure 2: LSTM-GAN architecture. The generator (red) uses stacked LSTM layers to denoise EEG, while the discriminator (blue) evaluates signal fidelity. Adapted from [Brophy et al., 2022](#).

4.2 3.2 Data Preprocessing and Artifact Masking

EEG signals in .edf format are processed using a bipolar montage. The continuous recordings are segmented into 4-second windows, normalized, and artifacts are identified using annotation labels (e.g., muscle, ocular, chewing). These masks enable the synthesis of noisy/clean pairs. For example, artifact masking is implemented as:

```
def mask_artifacts(eeg_segment, annotations):
    mask = np.zeros_like(eeg_segment)
    for ann in annotations:
        mask[ann.onset:ann.offset] = 1
    return mask
```

4.3 3.3 Model Training

The LSTM-GAN is trained by optimizing a composite loss that combines adversarial and mean squared error (MSE) losses:

1. Adversarial Loss:

$$\mathcal{L}_{adv} = E_{x \sim p_{data}(x)} [\log D(x)] + E_{z \sim p_z(z)} [\log(1 - D(G(z)))]$$

2. MSE Loss: To further align the denoised output with the clean EEG, we compute the MSE loss between the clean signal x and the generator output $G(x)$:

$$\mathcal{L}_{MSE} = E_{x \sim p_{data}(x)} [\|x - G(x)\|_2^2]$$

3. **Total Loss:** The total loss is a weighted sum of the adversarial and MSE losses:

$$\mathcal{L}_{total} = \lambda_{adv} \mathcal{L}_{adv} + \lambda_{MSE} \mathcal{L}_{MSE}$$

During training, the generator minimizes $\mathcal{L}_{total}$, producing denoised EEG that is both adversarially realistic and closely aligned with the clean signal. Figure 3 shows the training pipeline.

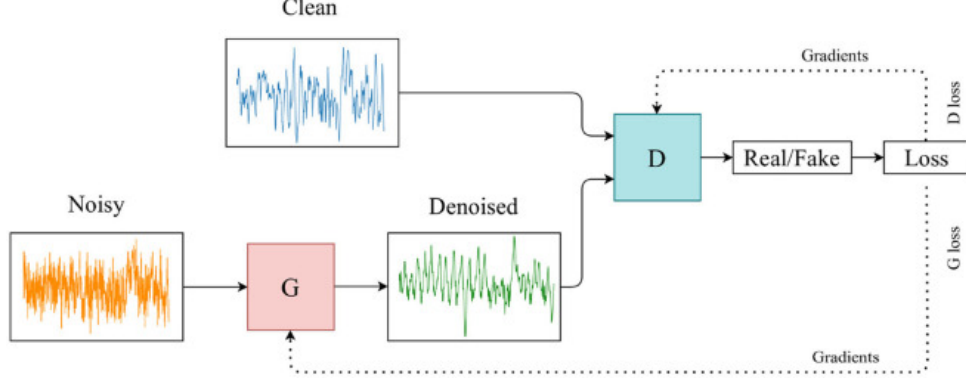


Figure 3: Training pipeline of the LSTM-GAN. The generator outputs denoised EEG (green) from noisy inputs (orange), while the discriminator (blue) classifies inputs as fake or real. Adapted from Brophy et al., 2022.

4.4 3.4 Inference Workflow

During inference, EEG recordings are:

- Segmented into 4-second windows.
- Preprocessed (normalized and padded as needed).
- Passed through the trained generator to produce denoised signals.
- Reassembled for evaluation using metrics such as SNR, RMSE, and Pearson correlation.

This inference process, implemented in PyTorch, ensures efficient and robust real-time artifact removal.

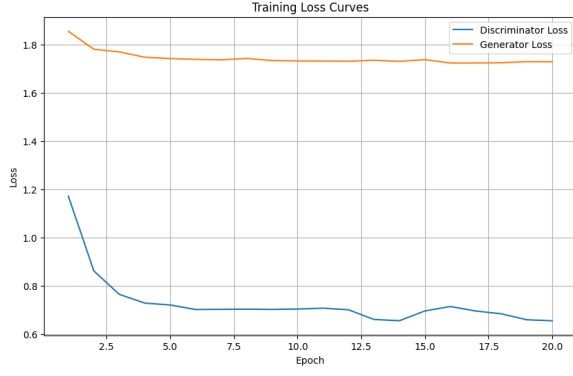
5 Results and Evaluation

Our experimental results illustrate the performance of the LSTM-GAN model through training/testing curves, a time-domain comparison, and quantitative metrics.

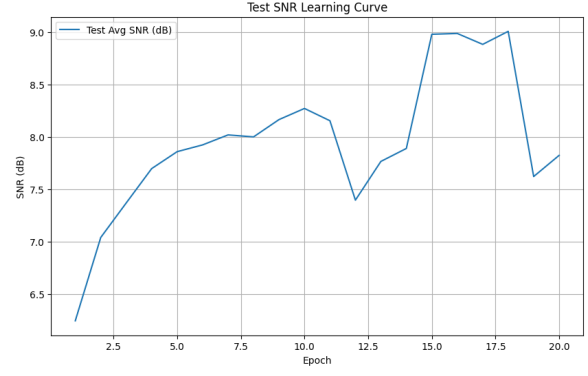
5.1 Training and Testing Performance

Figure 4a shows that both generator and discriminator losses drop sharply within the first four epochs, indicating rapid convergence before gradually stabilizing. Concurrently, the test SNR (Figure 4b) steadily rises to 8.3 dB by epoch 10, briefly dips at epoch 11, peaks at 9 dB by epoch 15, and then slightly falls by epoch 18—suggesting that the model overcomes early fluctuations to achieve higher signal clarity.

Similarly, the test RMSE (Figure 5a) decreases from 0.99 to 0.88 in the first 10 epochs, with minor variations thereafter. The Pearson correlation (Figure 5b) improves from 0.836 to a peak of 0.876 between epochs 15 and 17, further supporting the model’s enhanced ability to preserve signal fidelity.

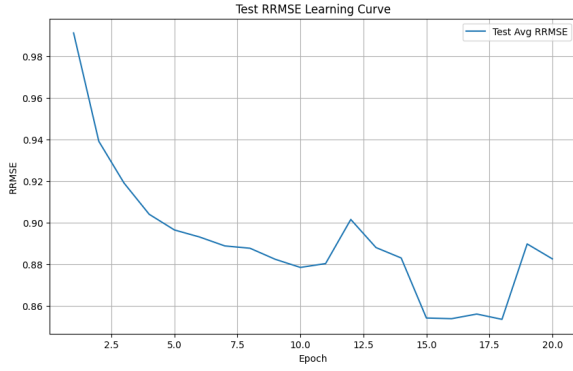


(a) Training Loss

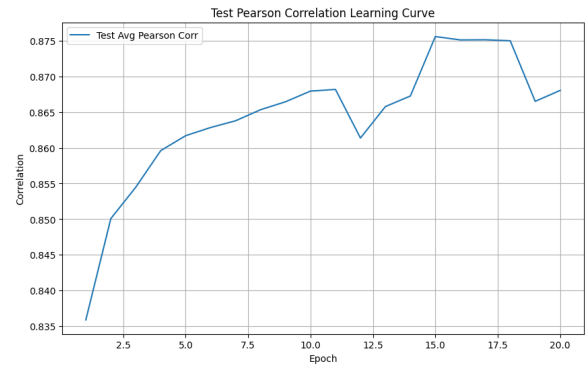


(b) Test SNR

Figure 4: Left: Training loss curves for the generator and discriminator, showing rapid decline and stabilization within the first 4 epochs. Right: Test SNR increases to 8.3 dB by epoch 10, dips at epoch 11, peaks at 9 dB at epoch 15, and then slightly decreases.



(a) Test RMSE



(b) Pearson Correlation

Figure 5: Left: Test RMSE decreases from 0.99 to 0.88 in the first 10 epochs, with minor fluctuations thereafter. Right: Pearson correlation increases from 0.836 to a peak of 0.876 between epochs 15–17, then slightly decreases.

5.2 Time-Domain Comparison

As shown in Figure 6, a visual comparison of the signals reveals:

- **Raw EEG with Artifacts (orange):** Exhibits significant noise.
- **Denoised EEG (blue):** Closely approximates the clean EEG.
- **Clean Reference (green):** Serves as the gold standard.

The near-perfect alignment of the blue (denoised) and green (clean reference) curves confirms that the model effectively removes artifacts while preserving the underlying EEG (Liu et al., 2021; Zhang & Liu, 2021).

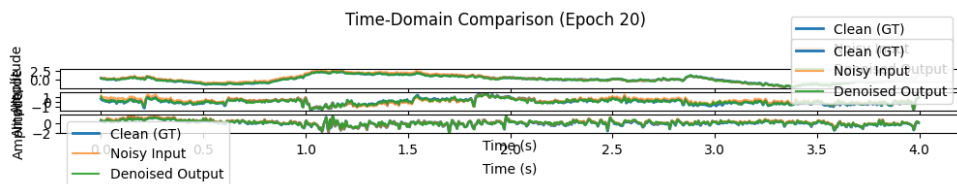


Figure 6: Time-domain comparison: raw EEG (orange), denoised EEG (blue), and clean reference (green).

5.3 Quantitative Metrics Evaluation

Our model’s performance is evaluated using several metrics:

- **SNR Improvement:** The average SNR increased to 14.39 dB (Std: 6.41 dB, 95% CI: [14.20, 14.59] dB). The relatively high standard deviation indicates significant variability—suggesting that while some EEG segments show strong enhancement in clarity, others may still retain residual artifacts.
- **RMSE Reduction:** The average RMSE decreased to 0.3174 (Std: 0.9321, 95% CI: [0.29, 0.34]). This low average error implies that the denoised EEG is very close to the clean reference, evidencing strong reconstruction performance.
- **Signal Fidelity (Pearson Correlation):** A correlation coefficient of 0.877 indicates a strong positive relationship between the denoised signal and the clean reference. This high correlation confirms that essential EEG structures are preserved, though minor differences remain, as a perfect correlation of 1 is not achieved.
- **Statistical Significance:** A paired t-test produced a t-statistic of 43.04 with a p-value below 0.0001. The high t-stat reflects a large and consistent difference between the noisy and denoised EEG, while the very small p-value confirms that this difference is highly unlikely to be due to random chance.

5.4 Overall Evaluation

The quantitative metrics and visual time-domain comparisons confirm that our LSTM-GAN model significantly improves EEG signal quality—evidenced by enhanced SNR, reduced RMSE, and strong Pearson correlation. However, the observed variability, particularly in cases with high-amplitude artifacts, indicates that while many segments are well denoised, some may still retain residual noise. Overall, these results endorse the model’s clinical potential for EEG analysis despite some remaining challenges.

6 Conclusion and Future Work

6.1 Conclusion

In this project, we developed an LSTM-based GAN for removing artifacts from clinical EEG recordings using the TUAR dataset. The approach demonstrates robust performance, achieving an average SNR improvement to 14.39 dB, RMSE reduction to 0.3174, and a Pearson correlation of 0.877 between the denoised and clean signals. Visual time-domain comparisons further confirm that the denoised output approximates the clean EEG closely, thus enhancing signal integrity for clinical analysis (Fatourehchi et al., 2007; Liu et al., 2021; Zhang & Liu, 2021). Nonetheless, performance variability in segments with high-amplitude artifacts remains an open challenge.

6.2 Future Work

Future efforts will primarily focus on:

- **Refining the Network Architecture:** Improve model robustness and data augmentation strategies to better handle high-amplitude artifacts.
- **Integrating Adaptive Techniques:** Incorporate complementary methods, such as adaptive filtering, to further reduce residual noise.

Code Availability

The complete implementation of this project is available online. You can access the code at: <https://www.kaggle.com/code/itzelxu/tuar-resgan>

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